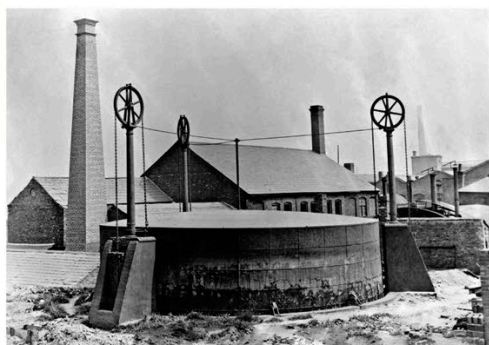


Powering up

Since the Industrial Revolution, engineers and industrialists have tried to generate power using different materials and machines. In turn, steam, gas, oil, and electricity have paved the way for many inventions. Indeed, breakthroughs in transportation, lighting, heat, and construction would have been impossible without these sources of power.



GAS POWER

The Scottish engineer William Murdoch worked in the mining area of Cornwall, UK, servicing steam engines. A by-product of heating coal is gas, and Murdoch figured out a way of capturing this gas in a tank (above) and igniting it. In 1792, Murdoch became the first person to light up a house (his own) using gas.

CRUDE OIL

Ancient peoples burned oil to generate light, but it was not until the mid-19th century that several individuals discovered how to extract oil from deep underground. Polish inventor Ignacy Łukasiewicz pioneered the oil industry as we know it when, in 1856, he created the world's first industrial oil refinery.



STEAM ENGINE

It is unlikely that English engineer Thomas Newcomen knew how important the steam engine would be when he invented it in 1710. His device was used for pumping water out of mines, and was later modified by James Watt (see pp.52–53), which led to steam locomotives. Steam powered the Industrial Revolution and changed the world.

Model of Newcomen's engine

Steam condenses in the cylinder and air pressure pushes the piston down.



Water is heated in the boiler and steam pushes a piston up.

The blue beam is attached to a red, curved end that looks like a bit like donkey's head.

▼ DRILLING FOR OIL

This oil-pumping unit is called a "nodding donkey" because of the way its driving beam swings up and down. It is in an oil field in Kazakhstan, Central Asia.